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```
%% Environmental Brain Cube - Suitability Analysis
% This script looks at one suitability run from the log file.
%
% It does these things:
% - reads the suitability log
% - picks one run
% - works with old or new log format
% - compares values to expected ranges
% - checks if values seem stable or drifting
% - makes plots for all main variables
% - gives a simple long term decision
```

```
clear; clc; close all;
```

```
%% settings
fileName = "ebc_suitability_log.csv";
runChoice = "3";     % "latest" or a run number like "2" (make sure run number is
accurate and quoted"")

showMinMaxBands = true;
showTrendLines = true;
useGroundTempInCore = true;

% prediction window settings
predictionMaxHours = 24;
predictionMinExtraFrac = 0.10;
predictionMaxExtraFrac = 0.80;

% settling OR trend behavior settings
prediction.recentFrac = 0.30;
prediction.startFrac = 0.20;
prediction.settleSlopeFrac = 0.35;
prediction.returnFrac = 0.25;
prediction.maxSettleHours = 8;
prediction.maxLinearHours = 24;
prediction.minPointsRecent = 6;

% prediction settings for battery / solar
prediction.minSlopeForLinear = 0.02;
prediction.energyMinSlopeForLinear = 0.05;
prediction.energySettleSlopeFrac = 0.20;

% curved prediction settings
prediction.allowCurved = true;
prediction.curveImproveFrac = 0.90;
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prediction.minCurveStrength = 0.002;
prediction.maxCurveAccelFrac = 0.80;
prediction.curvePoints = 140;

% keep graph future time under control
prediction.graphMaxHours = 24;

% power settings
powerCfg.measuredSystemCurrentA = 0.09;     % measured max system draw
powerCfg.measuredSupplyVoltageV = 5.83;     % measured supply voltage
powerCfg.solarRatedVoltageV = 5.0;          % panel rated voltage
powerCfg.solarRatedCurrentA = 0.20;         % panel rated current
powerCfg.solarRatedPowerW = powerCfg.solarRatedVoltageV *
powerCfg.solarRatedCurrentA;

% simple battery charge assumption
% this is just an estimate from available solar power
powerCfg.assumedBatteryVoltageV = 5.83;     % use measured supply side as simple
estimate
powerCfg.assumedBatteryCapacitymAh = 500;   % change if using a different battery
powerCfg.chargeEfficiency = 0.80;           % assumed charging loss estimate

% EXPECTED RANGES
% should match Arduino code
expected.TempAirMin = 20.0;
expected.TempAirMax = 25.0;

expected.TempGroundMin = 20.0;
expected.TempGroundMax = 25.0;

expected.HumMin = 40.0;
expected.HumMax = 55.0;

expected.PresMin = 950.0;
expected.PresMax = 1050.0;

expected.BattMin = 5.00;
expected.BattMax = 7.20;

expected.SolarMin = 3.50;
expected.SolarMax = 5.50;

expected.LightMin = 0;
expected.LightMax = 4095;

expected.AxMin = -16.0; expected.AxMax = 16.0;
expected.AyMin = -16.0; expected.AyMax = 16.0;
expected.AzMin = -16.0; expected.AzMax = 16.0;

expected.GxMin = -250.0; expected.GxMax = 250.0;
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```
expected.GyMin = -250.0; expected.GyMax = 250.0;
expected.GzMin = -250.0; expected.GzMax = 250.0;

%% long term decision settings
longTerm.centerGoodThreshold = 70;
longTerm.centerMarginalThreshold = 45;

longTerm.nearFutureHoursCritical = 24;
longTerm.nearFutureHoursWatch = 168;

longTerm.maxDriftTempC = 1.5;
longTerm.maxDriftHumPct = 5.0;
longTerm.maxDriftPreshPa = 3.0;
longTerm.maxDriftBattV = 0.25;
longTerm.maxDriftSolarV = 0.35;

longTerm.minDirectionalFraction = 0.70;
longTerm.windowFraction = 0.20;

%% check file
if ~isfile(fileName)
    error("File not found: " + fileName);
end

%% read file lines
txt = fileread(fileName);
lines = splitlines(string(txt));
lines = strtrim(lines);
lines = lines(lines ~= "");

%% keep only real run rows
dataLines = strings(0,1);

for i = 1:numel(lines)
    lineNow = lines(i);

    if startsWith(lineNow, "RUN") && ~startsWith(lineNow, "=== RUN")
        dataLines(end+1,1) = lineNow; %#ok<SAGROW>
    end
end

if isempty(dataLines)
    error("No valid RUN rows found.");
end

%% detect log format
firstParts = split(dataLines(1), ",");
numCols = numel(firstParts);

oldFormatCols = 19;
```

```
newFormatCols = 33;

if numCols == oldFormatCols
    logFormat = "old_single_value";
elseif numCols >= newFormatCols
    logFormat = "new_min_max";
else
    error("Unknown suitability log format. Found %d columns.", numCols);
end

fprintf("Detected log format: %s\n", logFormat);
```

```
Detected log format: new_min_max
```

```
%% create arrays
N = numel(dataLines);

runNum = nan(N,1);
timeSec = nan(N,1);

lightMin = nan(N,1); lightMax = nan(N,1); lightMid = nan(N,1); lightState =
strings(N,1);

bmeTempMin = nan(N,1); bmeTempMax = nan(N,1); bmeTempMid = nan(N,1);
bmeHumMin  = nan(N,1); bmeHumMax  = nan(N,1); bmeHumMid  = nan(N,1);
bmePresMin = nan(N,1); bmePresMax = nan(N,1); bmePresMid = nan(N,1);
dsTempMin  = nan(N,1); dsTempMax  = nan(N,1); dsTempMid  = nan(N,1);

battMin = nan(N,1); battMax = nan(N,1); battMid = nan(N,1);
solarMin = nan(N,1); solarMax = nan(N,1); solarMid = nan(N,1);

axMin = nan(N,1); axMax = nan(N,1); axMid = nan(N,1);
ayMin = nan(N,1); ayMax = nan(N,1); ayMid = nan(N,1);
azMin = nan(N,1); azMax = nan(N,1); azMid = nan(N,1);

gxMin = nan(N,1); gxMax = nan(N,1); gxMid = nan(N,1);
gyMin = nan(N,1); gyMax = nan(N,1); gyMid = nan(N,1);
gzMin = nan(N,1); gzMax = nan(N,1); gzMid = nan(N,1);

tempWarn = nan(N,1);
humWarn = nan(N,1);
presWarn = nan(N,1);

%% read rows
for i = 1:N
    p = split(dataLines(i), ",");
    p = strtrim(p);

    runNum(i) = sscanf(p(1), "RUN%d");
    timeSec(i) = toNum(p(3));
```

```
    if logFormat == "old_single_value"
        lightMid(i) = toNum(p(4));
        lightMin(i) = lightMid(i);
        lightMax(i) = lightMid(i);
        lightState(i) = classifyLight(lightMid(i), 1000, 3800);

        bmeTempMid(i) = toNum(p(5));  bmeTempMin(i) = bmeTempMid(i);  bmeTempMax(i)
= bmeTempMid(i);
        bmeHumMid(i)  = toNum(p(6));  bmeHumMin(i)  = bmeHumMid(i);   bmeHumMax(i)
= bmeHumMid(i);
        bmePresMid(i) = toNum(p(7));  bmePresMin(i) = bmePresMid(i);  bmePresMax(i)
= bmePresMid(i);
        dsTempMid(i)  = toNum(p(8));  dsTempMin(i)  = dsTempMid(i);   dsTempMax(i)
= dsTempMid(i);

        battMid(i)  = toNum(p(9));   battMin(i)  = battMid(i);   battMax(i)  =
battMid(i);
        solarMid(i) = toNum(p(10));  solarMin(i) = solarMid(i);  solarMax(i) =
solarMid(i);

        axMid(i) = toNum(p(11)); axMin(i) = axMid(i); axMax(i) = axMid(i);
        ayMid(i) = toNum(p(12)); ayMin(i) = ayMid(i); ayMax(i) = ayMid(i);
        azMid(i) = toNum(p(13)); azMin(i) = azMid(i); azMax(i) = azMid(i);

        gxMid(i) = toNum(p(14)); gxMin(i) = gxMid(i); gxMax(i) = gxMid(i);
        gyMid(i) = toNum(p(15)); gyMin(i) = gyMid(i); gyMax(i) = gyMid(i);
        gzMid(i) = toNum(p(16)); gzMin(i) = gzMid(i); gzMax(i) = gzMid(i);

        tempWarn(i) = toNum(p(17));
        humWarn(i) = toNum(p(18));
        presWarn(i) = toNum(p(19));

    else
        lightMin(i) = toNum(p(4));
        lightMax(i) = toNum(p(5));
        lightMid(i) = mean([lightMin(i), lightMax(i)], 'omitnan');
        lightState(i) = string(p(6));

        bmeTempMin(i) = toNum(p(7));   bmeTempMax(i) = toNum(p(8));   bmeTempMid(i)
= mean([bmeTempMin(i), bmeTempMax(i)], 'omitnan');
        bmeHumMin(i)  = toNum(p(9));   bmeHumMax(i)  = toNum(p(10));  bmeHumMid(i)
= mean([bmeHumMin(i), bmeHumMax(i)], 'omitnan');
        bmePresMin(i) = toNum(p(11));  bmePresMax(i) = toNum(p(12));  bmePresMid(i)
= mean([bmePresMin(i), bmePresMax(i)], 'omitnan');
        dsTempMin(i)  = toNum(p(13));  dsTempMax(i)  = toNum(p(14));  dsTempMid(i)
= mean([dsTempMin(i), dsTempMax(i)], 'omitnan');

        battMin(i) = toNum(p(15));   battMax(i) = toNum(p(16));   battMid(i) =
mean([battMin(i), battMax(i)], 'omitnan');
```

```
        solarMin(i) = toNum(p(17));  solarMax(i) = toNum(p(18));  solarMid(i) =
mean([solarMin(i), solarMax(i)], 'omitnan');

        axMin(i) = toNum(p(19)); axMax(i) = toNum(p(20)); axMid(i) =
mean([axMin(i), axMax(i)], 'omitnan');
        ayMin(i) = toNum(p(21)); ayMax(i) = toNum(p(22)); ayMid(i) =
mean([ayMin(i), ayMax(i)], 'omitnan');
        azMin(i) = toNum(p(23)); azMax(i) = toNum(p(24)); azMid(i) =
mean([azMin(i), azMax(i)], 'omitnan');

        gxMin(i) = toNum(p(25)); gxMax(i) = toNum(p(26)); gxMid(i) =
mean([gxMin(i), gxMax(i)], 'omitnan');
        gyMin(i) = toNum(p(27)); gyMax(i) = toNum(p(28)); gyMid(i) =
mean([gyMin(i), gyMax(i)], 'omitnan');
        gzMin(i) = toNum(p(29)); gzMax(i) = toNum(p(30)); gzMid(i) =
mean([gzMin(i), gzMax(i)], 'omitnan');

        tempWarn(i) = toNum(p(31));
        humWarn(i) = toNum(p(32));
        presWarn(i) = toNum(p(33));
    end
end

%% make table
T = table(runNum,timeSec, ...
    lightMin,lightMax,lightMid,lightState, ...
    bmeTempMin,bmeTempMax,bmeTempMid, ...
    bmeHumMin,bmeHumMax,bmeHumMid, ...
    bmePresMin,bmePresMax,bmePresMid, ...
    dsTempMin,dsTempMax,dsTempMid, ...
    battMin,battMax,battMid, ...
    solarMin,solarMax,solarMid, ...
    axMin,axMax,axMid, ...
    ayMin,ayMax,ayMid, ...
    azMin,azMax,azMid, ...
    gxMin,gxMax,gxMid, ...
    gyMin,gyMax,gyMid, ...
    gzMin,gzMax,gzMid, ...
    tempWarn,humWarn,presWarn);

%% choose run
runs = unique(T.runNum);
runs = runs(~isnan(runs));

if isempty(runs)
    error("No valid runs found.");
end

if runChoice == "latest"
    runToUse = max(runs);
```

```
else
    runToUse = double(runChoice);
end

D = T(T.runNum == runToUse, :);
D = sortrows(D, "timeSec");

if isempty(D)
    error("Selected run not found.");
end

fprintf("Using RUN%d\n", runToUse);
```

```
Using RUN3
```

```
fprintf("Rows logged: %d\n", height(D));
```

```
Rows logged: 248
```

```
fprintf("Run duration: %.2f s\n", max(D.timeSec));
```

```
Run duration: 500.01 s
```

```
%% average values
runDurationSec = max(D.timeSec);

avgAirTemp = mean(D.bmeTempMid, 'omitnan');
avgGroundTemp = mean(D.dsTempMid, 'omitnan');
avgHum = mean(D.bmeHumMid, 'omitnan');
avgPres = mean(D.bmePresMid, 'omitnan');
avgBatt = mean(D.battMid, 'omitnan');
avgSolar = mean(D.solarMid, 'omitnan');
avgLight = mean(D.lightMid, 'omitnan');

avgAx = mean(D.axMid, 'omitnan');
avgAy = mean(D.ayMid, 'omitnan');
avgAz = mean(D.azMid, 'omitnan');

avgGx = mean(D.gxMid, 'omitnan');
avgGy = mean(D.gyMid, 'omitnan');
avgGz = mean(D.gzMid, 'omitnan');

summaryStats = table( ...
    avgAirTemp, avgGroundTemp, avgHum, avgPres, avgBatt, avgSolar, avgLight, ...
    avgAx, avgAy, avgAz, avgGx, avgGy, avgGz, ...
    'VariableNames', {'AvgAirTemp','AvgGroundTemp','AvgHumidity','AvgPressure', ...

'AvgBattery','AvgSolar','AvgLight','AvgAx','AvgAy','AvgAz','AvgGx','AvgGy','AvgGz'})
;

disp("Run average values:")
```

Run average values:

```
disp(summaryStats)
```

AvgAirTemp	AvgGroundTemp	AvgHumidity	AvgPressure	AvgBattery	AvgSolar	AvgLight	AvgAx	A
22.782	20.447	46.284	1008.1	5.7183	4.7869	3781.9	0.31395	-0

```
%% compare averages to expected ranges
[airPass, airClose, airOffset] = compareToRange(avgAirTemp, expected.TempAirMin,
expected.TempAirMax);
[groundPass, groundClose, groundOffset] = compareToRange(avgGroundTemp,
expected.TempGroundMin, expected.TempGroundMax);
[humPass, humClose, humOffset] = compareToRange(avgHum, expected.HumMin,
expected.HumMax);
[presPass, presClose, presOffset] = compareToRange(avgPres, expected.PresMin,
expected.PresMax);

[battPass, battClose, battOffset] = compareToRange(avgBatt, expected.BattMin,
expected.BattMax);
[solarPass, solarClose, solarOffset] = compareToRange(avgSolar, expected.SolarMin,
expected.SolarMax);
[lightPass, lightClose, lightOffset] = compareToRange(avgLight, expected.LightMin,
expected.LightMax);

[axPass, axClose, axOffset] = compareToRange(avgAx, expected.AxMin, expected.AxMax);
[ayPass, ayClose, ayOffset] = compareToRange(avgAy, expected.AyMin, expected.AyMax);
[azPass, azClose, azOffset] = compareToRange(avgAz, expected.AzMin, expected.AzMax);

[gxPass, gxClose, gxOffset] = compareToRange(avgGx, expected.GxMin, expected.GxMax);
[gyPass, gyClose, gyOffset] = compareToRange(avgGy, expected.GyMin, expected.GyMax);
[gzPass, gzClose, gzOffset] = compareToRange(avgGz, expected.GzMin, expected.GzMax);

Parameter = [ ...
    "Air Temperature"; "Ground Temperature"; "Humidity"; "Pressure"; ...
    "Battery"; "Solar"; "Light"; ...
    "Accel X"; "Accel Y"; "Accel Z"; ...
    "Gyro X"; "Gyro Y"; "Gyro Z"];

RunAverage = [ ...
    avgAirTemp; avgGroundTemp; avgHum; avgPres; ...
    avgBatt; avgSolar; avgLight; ...
    avgAx; avgAy; avgAz; ...
    avgGx; avgGy; avgGz];

TargetMin = [ ...
    expected.TempAirMin; expected.TempGroundMin; expected.HumMin; expected.PresMin;
...
    expected.BattMin; expected.SolarMin; expected.LightMin; ...
```

```
    expected.AxMin; expected.AyMin; expected.AzMin; ...
    expected.GxMin; expected.GyMin; expected.GzMin];

TargetMax = [ ...
    expected.TempAirMax; expected.TempGroundMax; expected.HumMax; expected.PresMax;
...
    expected.BattMax; expected.SolarMax; expected.LightMax; ...
    expected.AxMax; expected.AyMax; expected.AzMax; ...
    expected.GxMax; expected.GyMax; expected.GzMax];

Pass = [ ...
    airPass; groundPass; humPass; presPass; ...
    battPass; solarPass; lightPass; ...
    axPass; ayPass; azPass; ...
    gxPass; gyPass; gzPass];

ClosenessPercent = [ ...
    airClose; groundClose; humClose; presClose; ...
    battClose; solarClose; lightClose; ...
    axClose; ayClose; azClose; ...
    gxClose; gyClose; gzClose];

CenterOffsetPercent = [ ...
    airOffset; groundOffset; humOffset; presOffset; ...
    battOffset; solarOffset; lightOffset; ...
    axOffset; ayOffset; azOffset; ...
    gxOffset; gyOffset; gzOffset];

ComparisonTable = table(Parameter, RunAverage, TargetMin, TargetMax, ...
    Pass, ClosenessPercent, CenterOffsetPercent);

disp("Comparison to expected ranges:")
```

```
Comparison to expected ranges:
```

```
disp(ComparisonTable)
```

Parameter	RunAverage	TargetMin	TargetMax	Pass	ClosenessPercent	CenterOffsetPercent
"Air Temperature"	22.782	20	25	true	88.716	11.284
"Ground Temperature"	20.447	20	25	true	17.868	82.132
"Humidity"	46.284	40	55	true	83.793	16.207
"Pressure"	1008.1	950	1050	true	83.735	16.265
"Battery"	5.7183	5	7.2	true	65.299	34.701
"Solar"	4.7869	3.5	5.5	true	71.315	28.685
"Light"	3781.9	0	4095	true	15.291	84.709
"Accel X"	0.31395	-16	16	true	98.038	1.9622
"Accel Y"	-0.71393	-16	16	true	95.538	4.4621
"Accel Z"	8.9636	-16	16	true	43.978	56.022
"Gyro X"	0.88518	-250	250	true	99.646	0.35407
"Gyro Y"	1.047	-250	250	true	99.581	0.41879
"Gyro Z"	-1.6077	-250	250	true	99.357	0.64307

```
%% trend and drift checks
timeHr = D.timeSec / 3600;

trend.airTemp = analyzeLongTermTrend(timeHr, D.bmeTempMid, expected.TempAirMin,
expected.TempAirMax, ...
    longTerm.maxDriftTempC, longTerm.minDirectionalFraction,
longTerm.windowFraction);

trend.groundTemp = analyzeLongTermTrend(timeHr, D.dsTempMid,
expected.TempGroundMin, expected.TempGroundMax, ...
    longTerm.maxDriftTempC, longTerm.minDirectionalFraction,
longTerm.windowFraction);

trend.humidity = analyzeLongTermTrend(timeHr, D.bmeHumMid, expected.HumMin,
expected.HumMax, ...
    longTerm.maxDriftHumPct, longTerm.minDirectionalFraction,
longTerm.windowFraction);

trend.pressure = analyzeLongTermTrend(timeHr, D.bmePresMid, expected.PresMin,
expected.PresMax, ...
    longTerm.maxDriftPreshPa, longTerm.minDirectionalFraction,
longTerm.windowFraction);

trend.battery = analyzeLongTermTrend(timeHr, D.battMid, expected.BattMin,
expected.BattMax, ...
    longTerm.maxDriftBattV, longTerm.minDirectionalFraction,
longTerm.windowFraction);

trend.solar = analyzeLongTermTrend(timeHr, D.solarMid, expected.SolarMin,
expected.SolarMax, ...
    longTerm.maxDriftSolarV, longTerm.minDirectionalFraction,
longTerm.windowFraction);

TrendSummary = table( ...
    ["Air Temp"; "Ground Temp"; "Humidity"; "Pressure"; "Battery"; "Solar"], ...
    [trend.airTemp.startEndDrift; trend.groundTemp.startEndDrift;
trend.humidity.startEndDrift; ...
     trend.pressure.startEndDrift; trend.battery.startEndDrift;
trend.solar.startEndDrift], ...
    [trend.airTemp.directionalFraction; trend.groundTemp.directionalFraction;
trend.humidity.directionalFraction; ...
     trend.pressure.directionalFraction; trend.battery.directionalFraction;
trend.solar.directionalFraction], ...
    [displayBoundaryTime(trend.airTemp); displayBoundaryTime(trend.groundTemp);
displayBoundaryTime(trend.humidity); ...
     displayBoundaryTime(trend.pressure); displayBoundaryTime(trend.battery);
displayBoundaryTime(trend.solar)], ...
    [string(trend.airTemp.behavior); string(trend.groundTemp.behavior);
string(trend.humidity.behavior); ...
```

```
     string(trend.pressure.behavior); string(trend.battery.behavior);
string(trend.solar.behavior)], ...
    'VariableNames',
{'Parameter','StartEndDrift','DirectionalFraction','TimeToBoundaryHours','Behavior'}
);

disp("Trend summary:")
```

```
Trend summary:
```

```
disp(TrendSummary)
```

Parameter	StartEndDrift	DirectionalFraction	TimeToBoundaryHours	Behavior
"Air Temp"	-0.8096	0.61943	NaN	"STABLE"
"Ground Temp"	-0.3235	0.19838	NaN	"STABLE"
"Humidity"	1.8919	0.53441	NaN	"STABLE"
"Pressure"	0.1113	0.43725	NaN	"STABLE"
"Battery"	-0.1685	0.36437	NaN	"STABLE"
"Solar"	-0.1539	0.40891	NaN	"STABLE"

```
%% power calculations
D.battPowerW = D.battMid * powerCfg.measuredSystemCurrentA;

avgBattPowerW = mean(D.battPowerW, 'omitnan');
maxBattPowerW = max(D.battPowerW, [], 'omitnan');
battEnergyWhRun = integratePowerWh(D.timeSec, D.battPowerW);

% direct measured supply reference
measuredSupplyPowerW = powerCfg.measuredSupplyVoltageV *
powerCfg.measuredSystemCurrentA;

% solar side estimate from rated panel power scaled by logged solar voltage
solarVoltageFrac = max(0, min(1.20, D.solarMid / powerCfg.solarRatedVoltageV));
D.solarPowerEstW = powerCfg.solarRatedPowerW * solarVoltageFrac;

avgSolarPowerEstW = mean(D.solarPowerEstW, 'omitnan');
maxSolarPowerEstW = max(D.solarPowerEstW, [], 'omitnan');
solarEnergyEstWhRun = integratePowerWh(D.timeSec, D.solarPowerEstW);

% net margin
D.netPowerMarginW = D.solarPowerEstW - D.battPowerW;
avgNetPowerMarginW = mean(D.netPowerMarginW, 'omitnan');

% 24 h predicted voltage and power
battPred24hV = predictValueAtHours(D.timeSec, D.battMid, expected.BattMin,
expected.BattMax, trend.battery, prediction, true, 24);
solarPred24hV = predictValueAtHours(D.timeSec, D.solarMid, expected.SolarMin,
expected.SolarMax, trend.solar, prediction, true, 24);

if ~isnan(battPred24hV)
```

```
    battPred24hPowerW = battPred24hV * powerCfg.measuredSystemCurrentA;
else
    battPred24hPowerW = NaN;
end

if ~isnan(solarPred24hV)
    solarPred24hPowerW = powerCfg.solarRatedPowerW * max(0, min(1.20,
solarPred24hV / powerCfg.solarRatedVoltageV));
else
    solarPred24hPowerW = NaN;
end

% battery voltage-drop check
battDropRateVph = min(0, trend.battery.recentSlopePerHour);

if isnan(battDropRateVph)
    battPowerDropAlignment = "NOT ENOUGH DATA";
elseif avgBattPowerW > 0 && battDropRateVph < 0
    battPowerDropAlignment = "ALIGNS";
elseif avgBattPowerW > 0 && abs(battDropRateVph) < 1e-6
    battPowerDropAlignment = "MOSTLY STABLE";
else
    battPowerDropAlignment = "MIXED";
end

% solar trend check
solarSlopeVph = trend.solar.recentSlopePerHour;

if isnan(solarSlopeVph)
    solarPowerTrendAlignment = "NOT ENOUGH DATA";
elseif avgSolarPowerEstW > avgBattPowerW && solarSlopeVph >= 0
    solarPowerTrendAlignment = "ALIGNS";
elseif avgSolarPowerEstW > avgBattPowerW && solarSlopeVph < 0
    solarPowerTrendAlignment = "HIGH BUT DROPPING";
elseif avgSolarPowerEstW <= avgBattPowerW && solarSlopeVph < 0
    solarPowerTrendAlignment = "ALIGNS";
else
    solarPowerTrendAlignment = "MIXED";
end

% battery charging estimate from solar surplus
% this is an assumed estimate, not a full battery model
batteryEnergyWhAssumed = (powerCfg.assumedBatteryCapacitymAh / 1000) *
powerCfg.assumedBatteryVoltageV;

if avgNetPowerMarginW > 0
    chargeTimeHoursIdeal = batteryEnergyWhAssumed / avgNetPowerMarginW;
    chargeTimeHoursAdj = batteryEnergyWhAssumed / (avgNetPowerMarginW *
powerCfg.chargeEfficiency);
else
```

```
    chargeTimeHoursIdeal = Inf;
    chargeTimeHoursAdj = Inf;
end

PowerSummary = table( ...
    avgBattPowerW, maxBattPowerW, battEnergyWhRun, ...
    avgSolarPowerEstW, maxSolarPowerEstW, solarEnergyEstWhRun, ...
    avgNetPowerMarginW, measuredSupplyPowerW, ...
    battPred24hV, battPred24hPowerW, ...
    solarPred24hV, solarPred24hPowerW, ...
    chargeTimeHoursIdeal, chargeTimeHoursAdj, ...
    string(battPowerDropAlignment), string(solarPowerTrendAlignment), ...
    'VariableNames', {'AvgBattPowerW','MaxBattPowerW','BattEnergyWhRun', ...
    'AvgSolarPowerEstW','MaxSolarPowerEstW','SolarEnergyEstWhRun', ...
    'AvgNetPowerMarginW','MeasuredSupplyPowerW', ...
    'BattPred24hV','BattPred24hPowerW', ...
    'SolarPred24hV','SolarPred24hPowerW', ...
    'ChargeTimeHrIdeal','ChargeTimeHrAdj', ...
    'BattPowerVsDrop','SolarPowerVsTrend'});

disp("Power summary:")
```

```
Power summary:
```

```
disp(PowerSummary)
```

AvgBattPowerW	MaxBattPowerW	BattEnergyWhRun	AvgSolarPowerEstW	MaxSolarPowerEstW	SolarEnergyEstWhR
0.51465	0.5418	0.071257	0.95737	0.996	0.13259

```
%% air temp vs ground temp
avgAirGroundDelta = mean(D.bmeTempMid - D.dsTempMid, 'omitnan');
maxAirGroundDelta = max(abs(D.bmeTempMid - D.dsTempMid), [], 'omitnan');

if nnz(~isnan(D.bmeTempMid) & ~isnan(D.dsTempMid)) >= 3
    airGroundCorr = corr(D.bmeTempMid, D.dsTempMid, 'Rows', 'complete');
else
    airGroundCorr = NaN;
end

ExtraSummary = table( ...
    avgAirGroundDelta, maxAirGroundDelta, airGroundCorr, ...
    'VariableNames',
{'AvgAirMinusGroundTempC','MaxAbsAirGroundTempDiffC','AirGroundCorrelation'});

disp("Extra deductions:")
```

```
Extra deductions:
```

```
disp(ExtraSummary)
```

AvgAirMinusGroundTempC	MaxAbsAirGroundTempDiffC	AirGroundCorrelation
2.3354	2.86	0.95298

```
%% long-term score
coreClosenessList = [airClose, humClose, presClose];

if useGroundTempInCore
    coreClosenessList = [coreClosenessList, groundClose];
end

coreAverageCloseness = mean(coreClosenessList, 'omitnan');

coreTrendPenalty = 0;
coreComments = strings(0,1);

[coreTrendPenalty, coreComments] = applyRealDriftPenalty( ...
    coreTrendPenalty, coreComments, "Air temperature", trend.airTemp, ...
    longTerm.nearFutureHoursCritical, longTerm.nearFutureHoursWatch);

if useGroundTempInCore
    [coreTrendPenalty, coreComments] = applyRealDriftPenalty( ...
        coreTrendPenalty, coreComments, "Ground temperature", trend.groundTemp, ...
        longTerm.nearFutureHoursCritical, longTerm.nearFutureHoursWatch);
end

[coreTrendPenalty, coreComments] = applyRealDriftPenalty( ...
    coreTrendPenalty, coreComments, "Humidity", trend.humidity, ...
    longTerm.nearFutureHoursCritical, longTerm.nearFutureHoursWatch);

[coreTrendPenalty, coreComments] = applyRealDriftPenalty( ...
    coreTrendPenalty, coreComments, "Pressure", trend.pressure, ...
    longTerm.nearFutureHoursCritical, longTerm.nearFutureHoursWatch);

longTermScore = max(0, coreAverageCloseness - coreTrendPenalty);
supportAverageCloseness = mean([battClose, solarClose, lightClose], 'omitnan');

coreFail = strcmp(trend.airTemp.behavior, "UNSAFE_RISING") || ...
           strcmp(trend.humidity.behavior, "UNSAFE_RISING") || ...
           strcmp(trend.pressure.behavior, "UNSAFE_RISING");

if useGroundTempInCore
    coreFail = coreFail || strcmp(trend.groundTemp.behavior, "UNSAFE_RISING");
end

if coreFail
    longTermDecision = "NOT IDEAL FOR LONG-TERM USE";
elseif longTermScore >= longTerm.centerGoodThreshold
    longTermDecision = "LIKELY SUITABLE LONG-TERM";
```

```
elseif longTermScore >= longTerm.centerMarginalThreshold
    longTermDecision = "MARGINALLY SUITABLE LONG-TERM";
else
    longTermDecision = "WATCH FOR NEAR-FUTURE ENVIRONMENTAL CHANGE";
end

%% warning percentages
tempWarnFraction = mean(D.tempWarn == 1, 'omitnan') * 100;
humWarnFraction = mean(D.humWarn == 1, 'omitnan') * 100;
presWarnFraction = mean(D.presWarn == 1, 'omitnan') * 100;

%% overall summary
OverallSummary = table( ...
    runToUse, runDurationSec, ...
    coreAverageCloseness, supportAverageCloseness, coreTrendPenalty, longTermScore,
...
    tempWarnFraction, humWarnFraction, presWarnFraction, ...
    string(longTermDecision), ...
    'VariableNames',
{'Run','DurationSec','CoreAverageClosenessPct','SupportAverageClosenessPct', ...

'CoreTrendPenalty','LongTermScorePct','TempWarnPct','HumWarnPct','PresWarnPct','Long
TermDecision'});

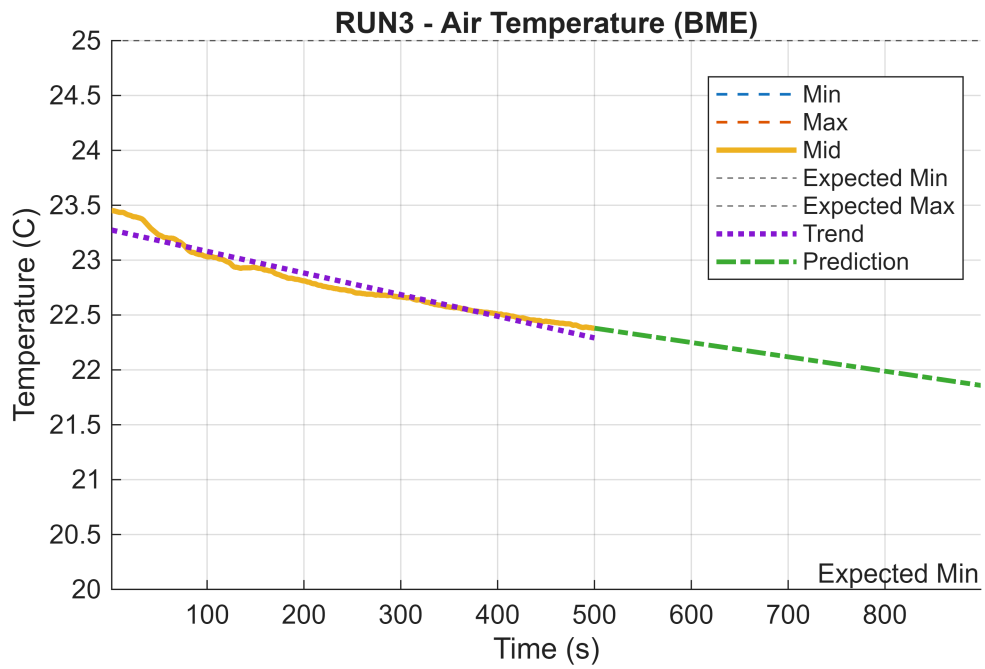
disp("Overall suitability summary:")
```

```
Overall suitability summary:
```

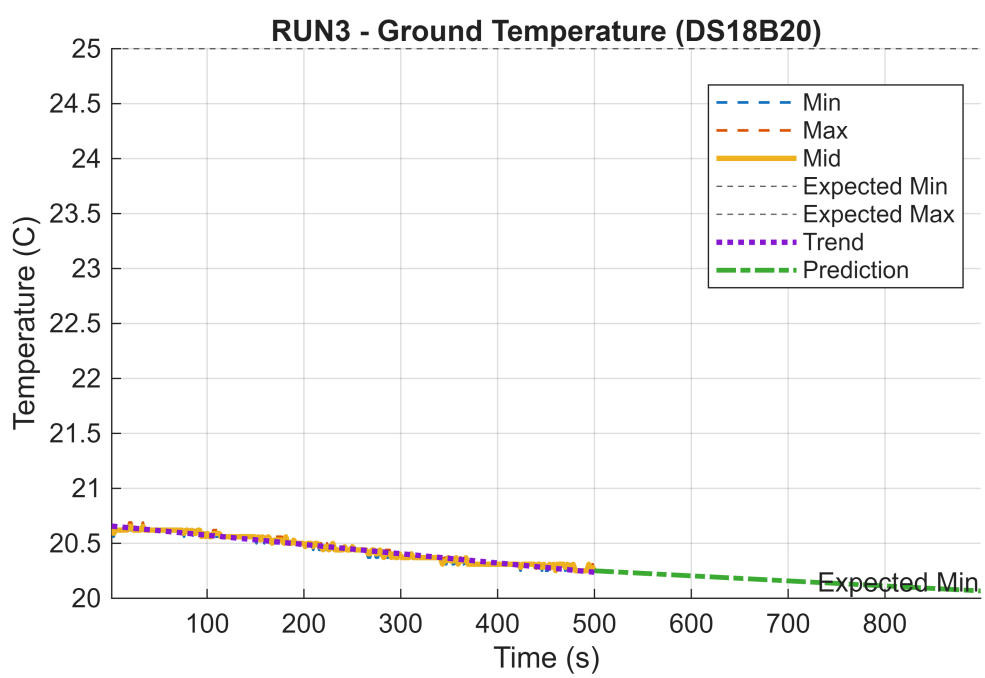
```
disp(OverallSummary)
```

Run	DurationSec	CoreAverageClosenessPct	SupportAverageClosenessPct	CoreTrendPenalty	LongTermScore
3	500.01	68.528	50.635	0	68.528

```
%% plots
plotVariableWithBand(D.timeSec, D.bmeTempMin, D.bmeTempMax, D.bmeTempMid, ...
    expected.TempAirMin, expected.TempAirMax, ...
    "Air Temperature (BME)", "Temperature (C)", runToUse, ...
    showMinMaxBands, showTrendLines, trend.airTemp, prediction,
predictionMinExtraFrac, predictionMaxExtraFrac, false);
```

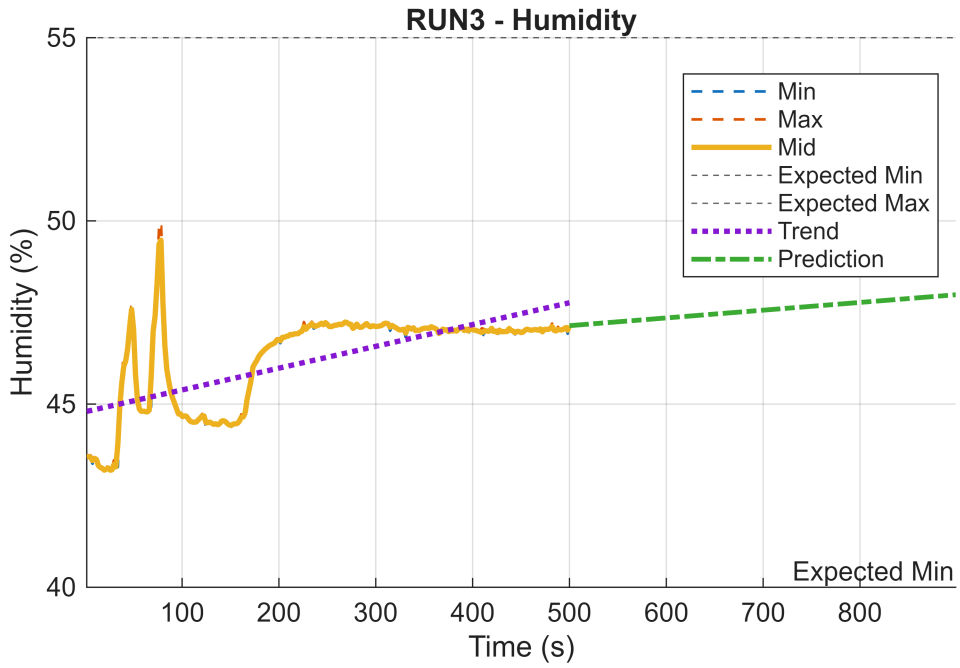



```
plotVariableWithBand(D.timeSec, D.dsTempMin, D.dsTempMax, D.dsTempMid, ...
    expected.TempGroundMin, expected.TempGroundMax, ...
    "Ground Temperature (DS18B20)", "Temperature (C)", runToUse, ...
    showMinMaxBands, showTrendLines, trend.groundTemp, prediction,
predictionMinExtraFrac, predictionMaxExtraFrac, false);
```

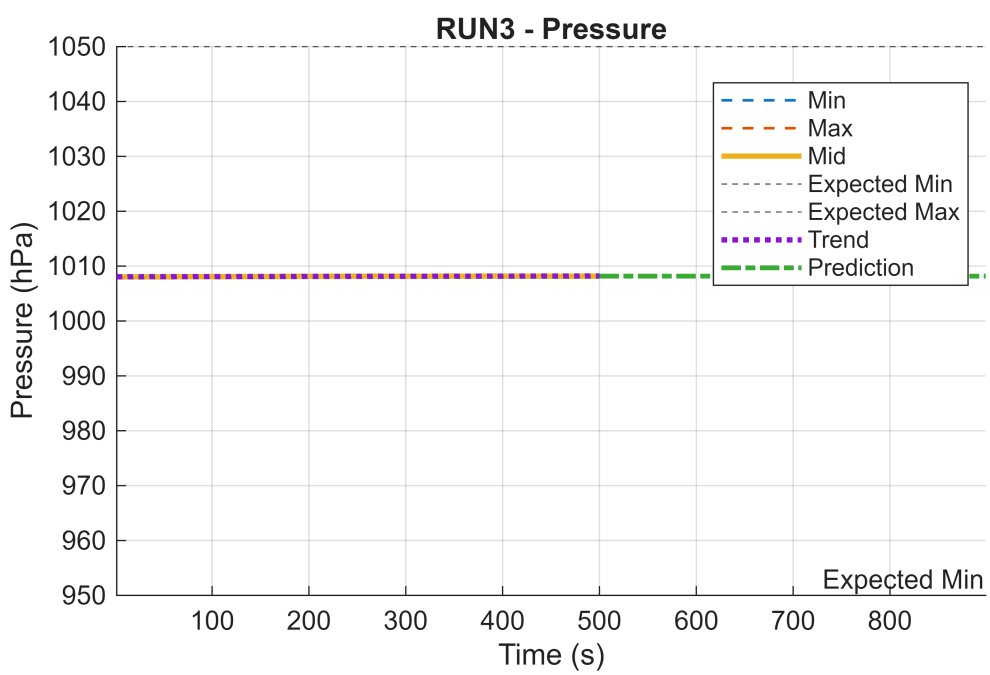



```
plotVariableWithBand(D.timeSec, D.bmeHumMin, D.bmeHumMax, D.bmeHumMid, ...
    expected.HumMin, expected.HumMax, ...
    "Humidity", "Humidity (%)", runToUse, ...
```

```
    showMinMaxBands, showTrendLines, trend.humidity, prediction,
predictionMinExtraFrac, predictionMaxExtraFrac, false);
```

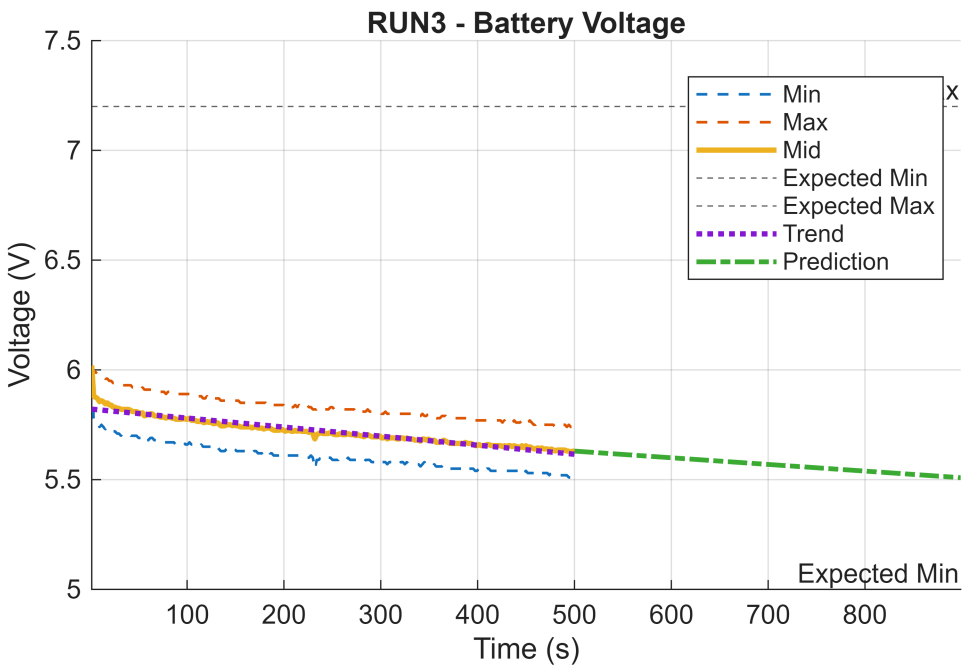



```
plotVariableWithBand(D.timeSec, D.bmePresMin, D.bmePresMax, D.bmePresMid, ...
    expected.PresMin, expected.PresMax, ...
    "Pressure", "Pressure (hPa)", runToUse, ...
    showMinMaxBands, showTrendLines, trend.pressure, prediction,
predictionMinExtraFrac, predictionMaxExtraFrac, false);
```

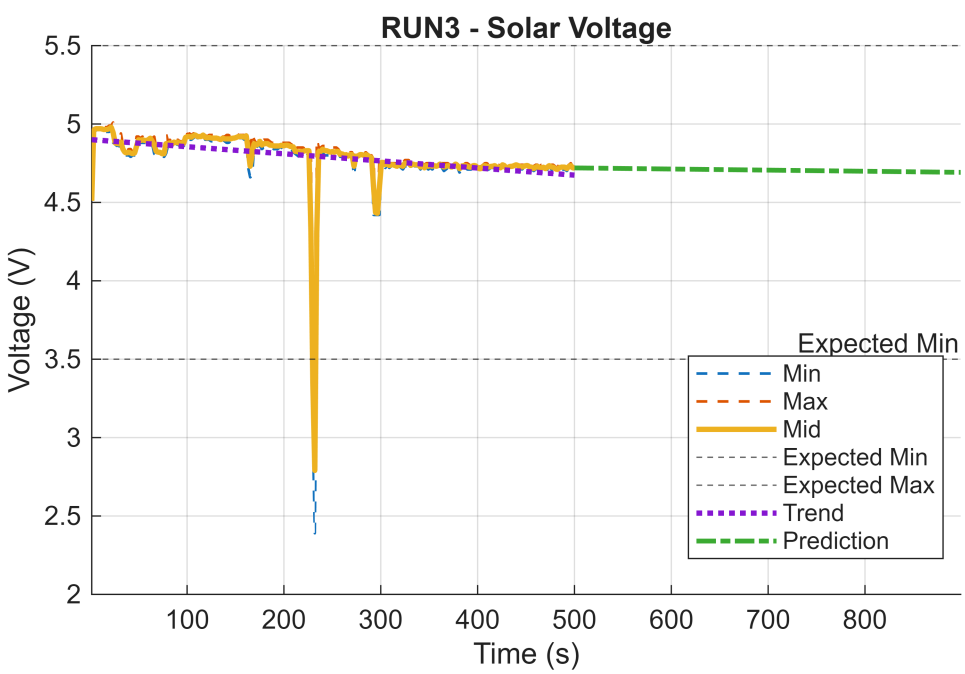



```
plotVariableWithBand(D.timeSec, D.battMin, D.battMax, D.battMid, ...
```

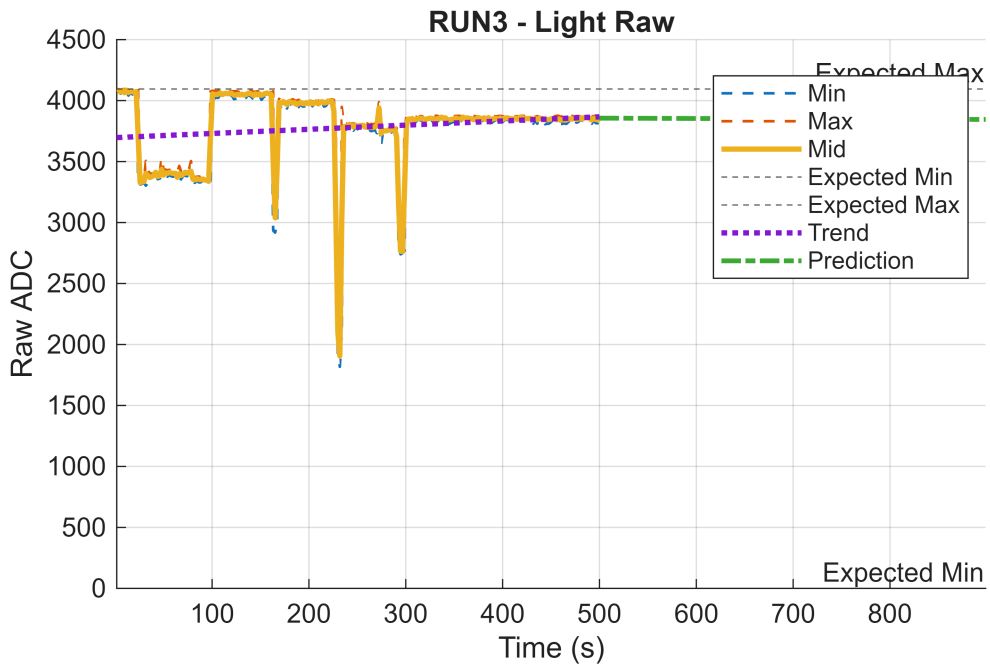
```
    expected.BattMin, expected.BattMax, ...
    "Battery Voltage", "Voltage (V)", runToUse, ...
    showMinMaxBands, showTrendLines, trend.battery, prediction,
predictionMinExtraFrac, predictionMaxExtraFrac, true);
```



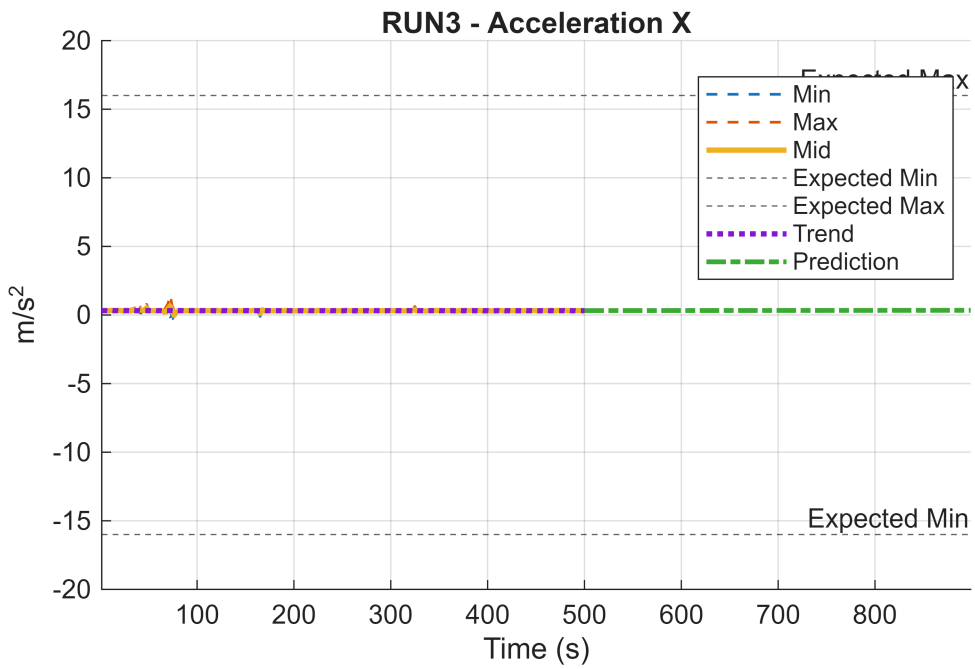

```
plotVariableWithBand(D.timeSec, D.solarMin, D.solarMax, D.solarMid, ...
    expected.SolarMin, expected.SolarMax, ...
    "Solar Voltage", "Voltage (V)", runToUse, ...
    showMinMaxBands, showTrendLines, trend.solar, prediction,
predictionMinExtraFrac, predictionMaxExtraFrac, true);
```



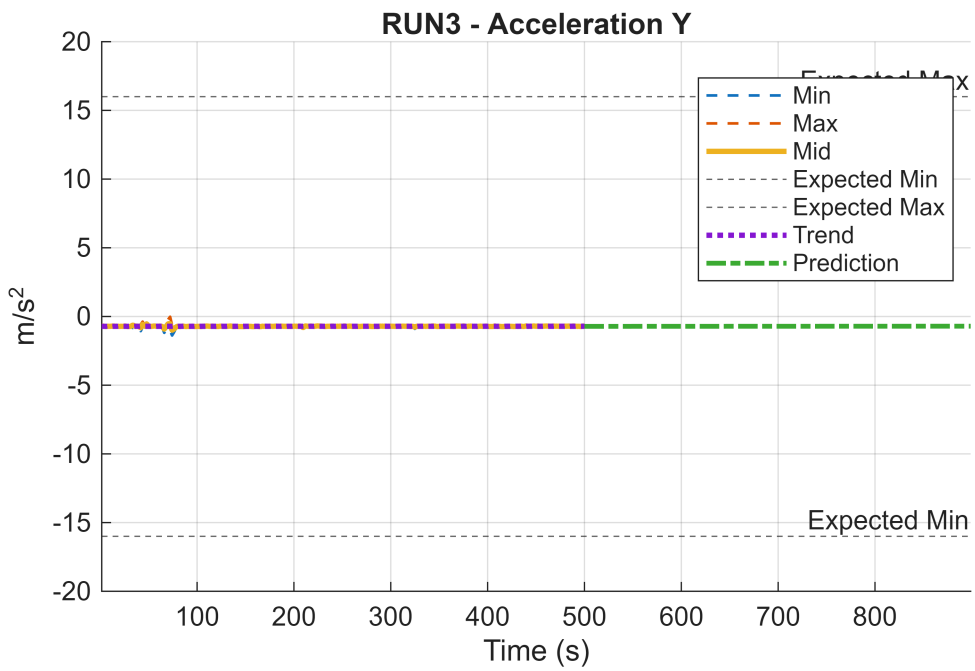
```
plotVariableWithBand(D.timeSec, D.lightMin, D.lightMax, D.lightMid, ...
    expected.LightMin, expected.LightMax, ...
    "Light Raw", "Raw ADC", runToUse, ...
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```




```
plotVariableWithBand(D.timeSec, D.axMin, D.axMax, D.axMid, ...
    expected.AxMin, expected.AxMax, ...
    "Acceleration X", "m/s^2", runToUse, ...
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```

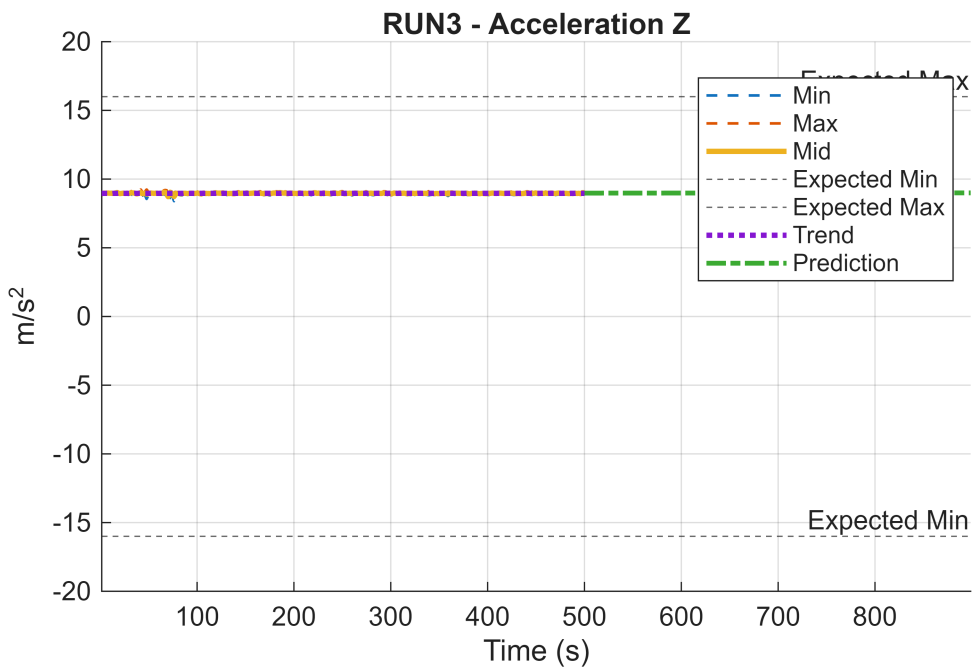



```
plotVariableWithBand(D.timeSec, D.ayMin, D.ayMax, D.ayMid, ...
    expected.AyMin, expected.AyMax, ...
    "Acceleration Y", "m/s^2", runToUse, ...
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```

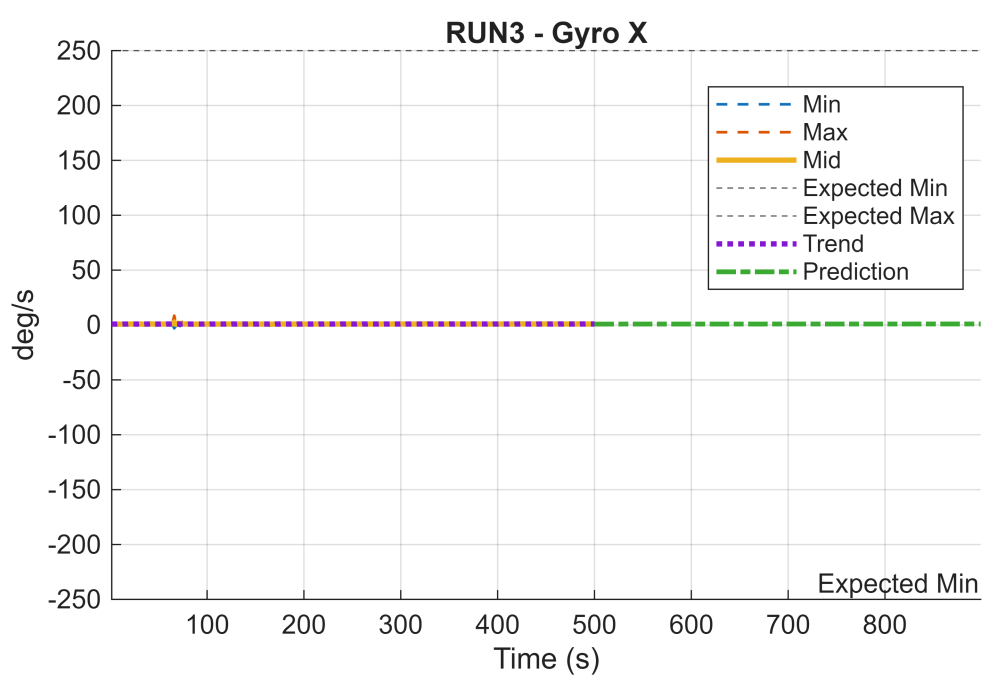



```
plotVariableWithBand(D.timeSec, D.azMin, D.azMax, D.azMid, ...
    expected.AzMin, expected.AzMax, ...
    "Acceleration Z", "m/s^2", runToUse, ...
```

```
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```

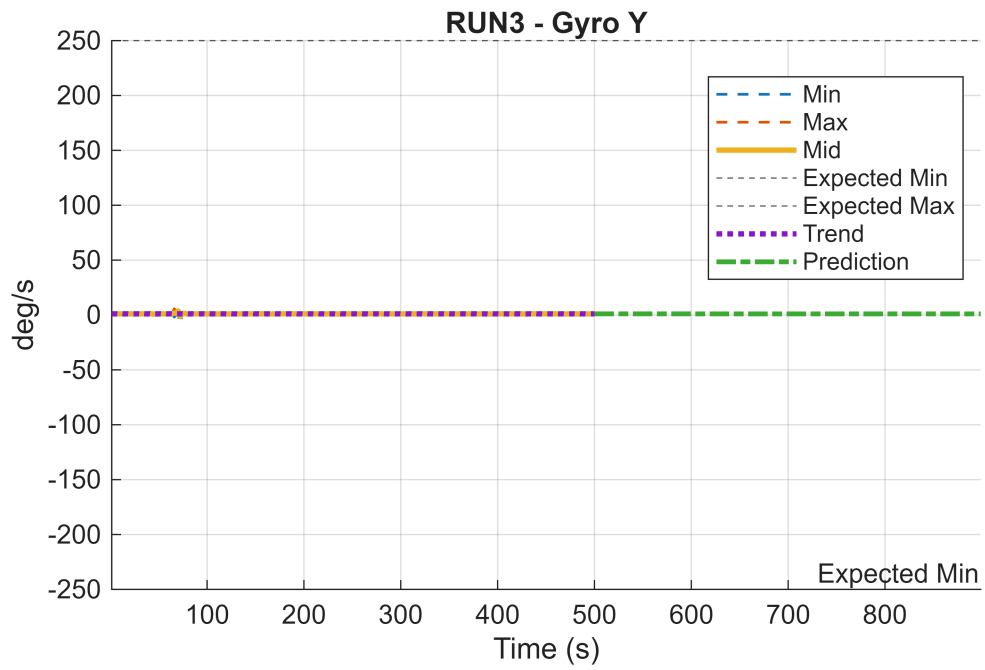



```
plotVariableWithBand(D.timeSec, D.gxMin, D.gxMax, D.gxMid, ...
    expected.GxMin, expected.GxMax, ...
    "Gyro X", "deg/s", runToUse, ...
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```

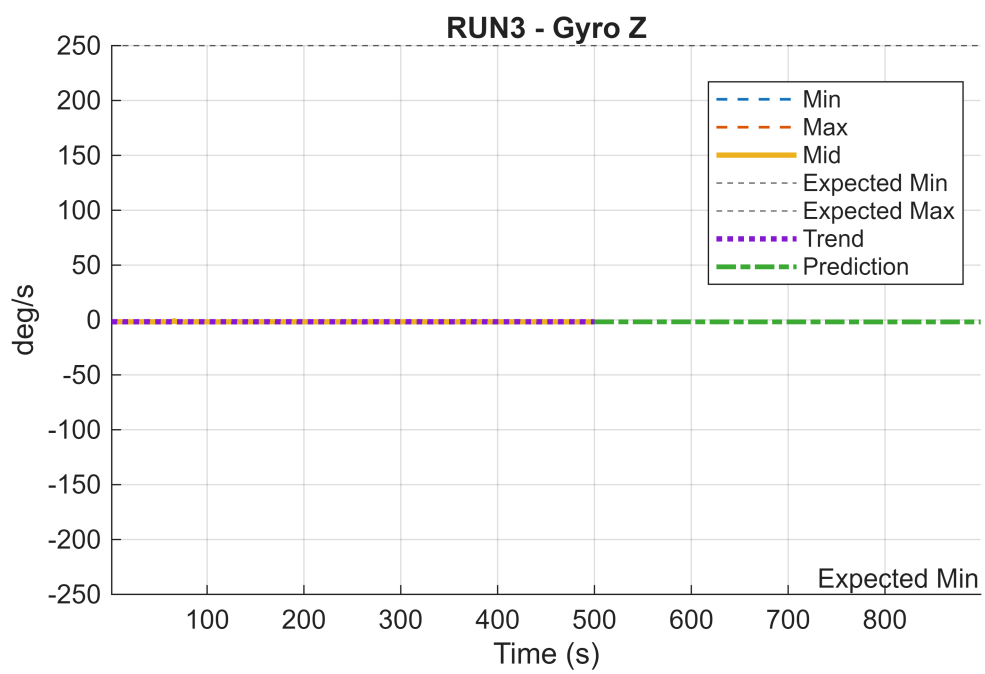



```
plotVariableWithBand(D.timeSec, D.gyMin, D.gyMax, D.gyMid, ...
```

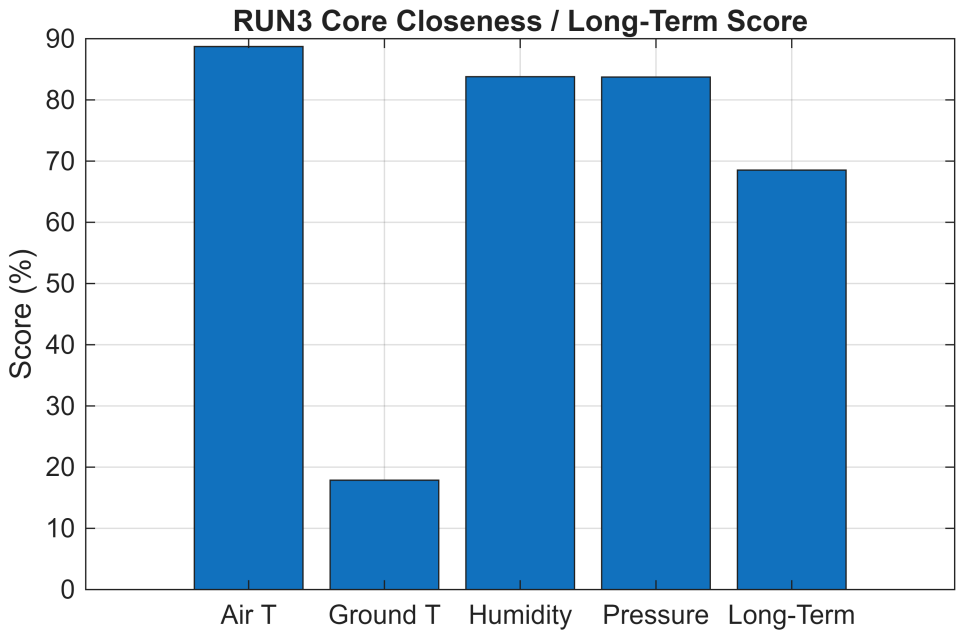
```
    expected.GyMin, expected.GyMax, ...
    "Gyro Y", "deg/s", runToUse, ...
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```



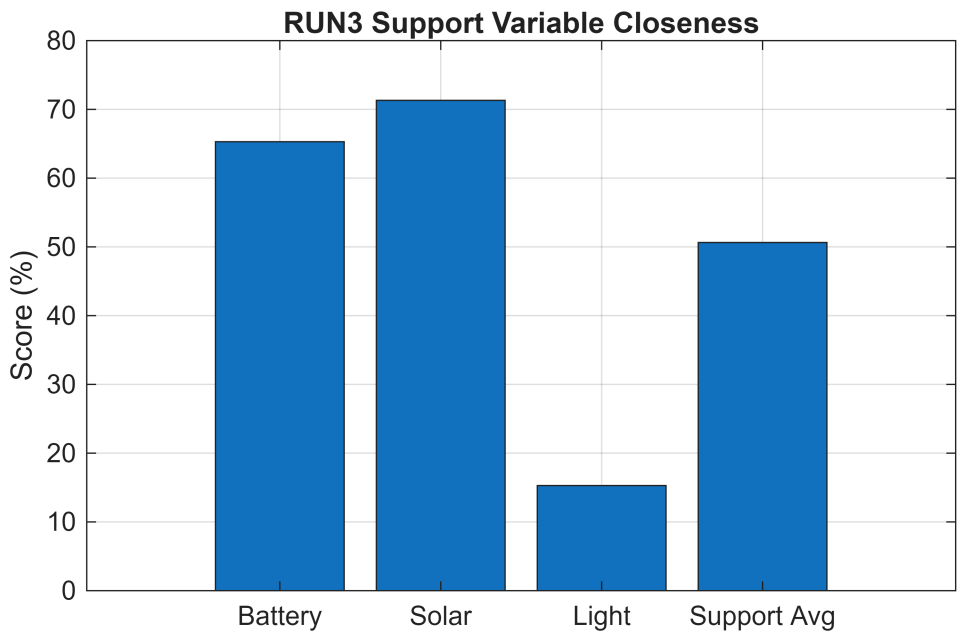

```
plotVariableWithBand(D.timeSec, D.gzMin, D.gzMax, D.gzMid, ...
    expected.GzMin, expected.GzMax, ...
    "Gyro Z", "deg/s", runToUse, ...
    showMinMaxBands, showTrendLines, struct(), prediction, predictionMinExtraFrac,
predictionMaxExtraFrac, false);
```



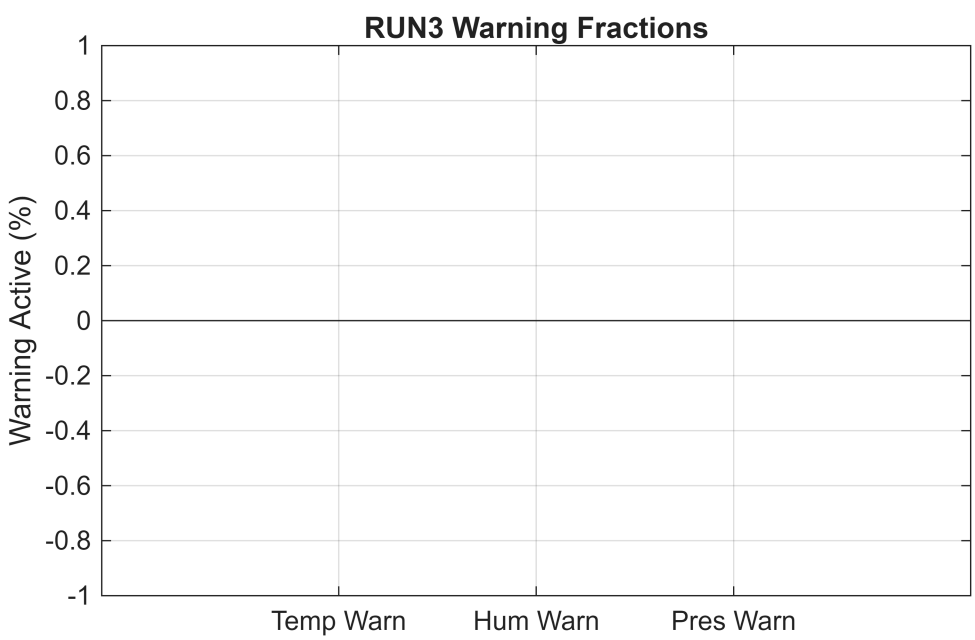
```
figure
bar([airClose, groundClose, humClose, presClose, longTermScore])
set(gca, 'XTickLabel', {'Air T','Ground T','Humidity','Pressure','Long-Term'})
ylabel('Score (%)')
title("RUN" + runToUse + " Core Closeness / Long-Term Score")
grid on
```




```
figure
bar([battClose, solarClose, lightClose, supportAverageCloseness])
set(gca, 'XTickLabel', {'Battery','Solar','Light','Support Avg'})
ylabel('Score (%)')
title("RUN" + runToUse + " Support Variable Closeness")
grid on
```

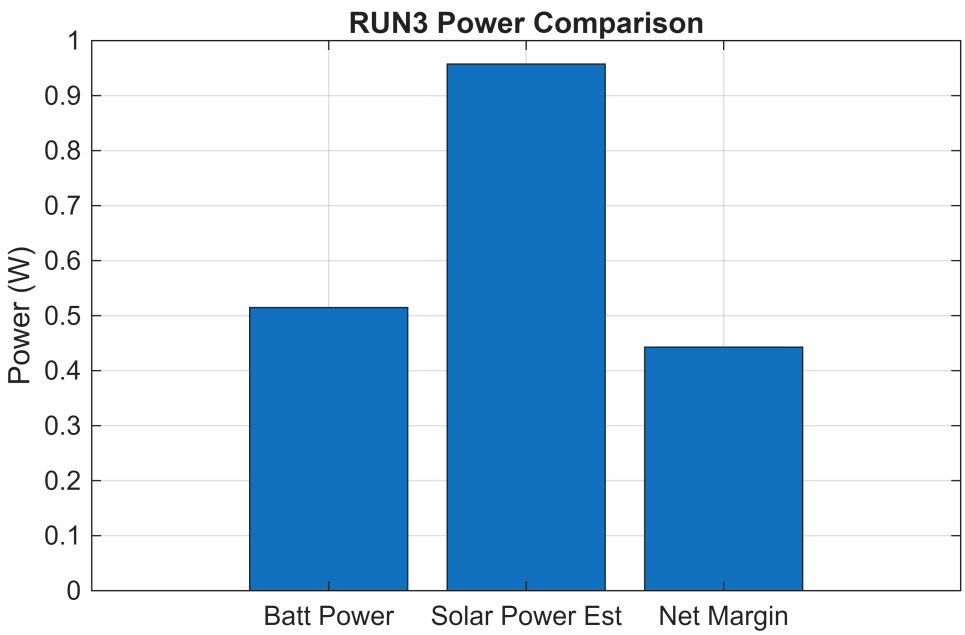



```
figure
bar([tempWarnFraction, humWarnFraction, presWarnFraction])
set(gca, 'XTickLabel', {'Temp Warn','Hum Warn','Pres Warn'})
ylabel('Warning Active (%)')
title("RUN" + runToUse + " Warning Fractions")
grid on
```




```
figure
bar([avgBattPowerW, avgSolarPowerEstW, avgNetPowerMarginW])
```

```
set(gca, 'XTickLabel', {'Batt Power','Solar Power Est','Net Margin'})
ylabel('Power (W)')
title("RUN" + runToUse + " Power Comparison")
grid on
```




```
%% conclusions
fprintf('\n============= SUITABILITY ANALYSIS =============\n');
```

```
============= SUITABILITY ANALYSIS =============
```

```
fprintf('Run: RUN%d | Duration: %.2f s | Format: %s\n', runToUse, runDurationSec,
logFormat);
```

```
Run: RUN3 | Duration: 500.01 s | Format: new_min_max
```

```
fprintf('\nAverages:\n');
```

```
Averages:
```

```
fprintf('- Air: %.2f C | Ground: %.2f C | Humidity: %.2f %% | Pressure: %.2f
hPa\n', ...
    avgAirTemp, avgGroundTemp, avgHum, avgPres);
```

```
- Air: 22.78 C | Ground: 20.45 C | Humidity: 46.28 % | Pressure: 1008.13 hPa
```

```
fprintf('- Battery: %.2f V | Solar: %.2f V | Light: %.2f\n', ...
    avgBatt, avgSolar, avgLight);
```

```
- Battery: 5.72 V | Solar: 4.79 V | Light: 3781.91
```

```
fprintf('\nCore check:\n');
```

```
Core check:
```

```
fprintf('- Air %.1f%% | Ground %.1f%% | Humidity %.1f%% | Pressure %.1f%%\n', ...
    airClose, groundClose, humClose, presClose);
```

```
- Air 88.7% | Ground 17.9% | Humidity 83.8% | Pressure 83.7%
```

```
fprintf('\nPower check:\n');
```

```
Power check:
```

```
fprintf('- Battery avg power: %.3f W\n', avgBattPowerW);
```

```
- Battery avg power: 0.515 W
```

```
fprintf('- Solar avg power est: %.3f W\n', avgSolarPowerEstW);
```

```
- Solar avg power est: 0.957 W
```

```
fprintf('- Net margin: %.3f W\n', avgNetPowerMarginW);
```

```
- Net margin: 0.443 W
```

```
fprintf('- Battery power vs drop: %s\n', battPowerDropAlignment);
```

```
- Battery power vs drop: ALIGNS
```

```
fprintf('- Solar power vs trend: %s\n', solarPowerTrendAlignment);
```

```
- Solar power vs trend: HIGH BUT DROPPING
```

```
if isfinite(chargeTimeHoursAdj)
    fprintf('- Est. battery charge time from solar: %.2f hr ideal, %.2f hr
adjusted\n', ...
        chargeTimeHoursIdeal, chargeTimeHoursAdj);
else
    fprintf('- Est. battery charge time from solar: not supported by current
average solar margin\n');
end
```

```
- Est. battery charge time from solar: 6.58 hr ideal, 8.23 hr adjusted
```

```
fprintf('\nTrend notes:\n');
```

```
Trend notes:
```

```
printTrend("Air", trend.airTemp);
```

```
- Air: drift = -0.810, directional consistency = 0.62, behavior = STABLE | no boundary estimate needed
```

```
printTrend("Ground", trend.groundTemp);
```

```
- Ground: drift = -0.324, directional consistency = 0.20, behavior = STABLE | no boundary estimate needed
```

```
printTrend("Humidity", trend.humidity);
```

```
- Humidity: drift = 1.892, directional consistency = 0.53, behavior = STABLE | no boundary estimate needed
```

```
printTrend("Pressure", trend.pressure);
```

```
- Pressure: drift = 0.111, directional consistency = 0.44, behavior = STABLE | no boundary estimate needed
```

```
printTrend("Battery", trend.battery);
```

```
- Battery: drift = -0.169, directional consistency = 0.36, behavior = STABLE | no boundary estimate needed
```

```
printTrend("Solar", trend.solar);
```

```
- Solar: drift = -0.154, directional consistency = 0.41, behavior = STABLE | no boundary estimate needed
```

```

fprintf('\nOther notes:\n');
```

```
Other notes:
```

```
fprintf('- Avg air-ground temp diff: %.3f C\n', avgAirGroundDelta);
```

```
- Avg air-ground temp diff: 2.335 C
```

```
if ~isnan(airGroundCorr)
    fprintf('- Air-ground temp correlation: %.3f\n', airGroundCorr);
end
```

```
- Air-ground temp correlation: 0.953
```

```

fprintf('\nResult:\n');
```

```
Result:
```

```
fprintf('- Long-term score: %.1f %%\n', longTermScore);
```

```
- Long-term score: 68.5 %
```

```
fprintf('- Decision: %s\n', longTermDecision);
```

```
- Decision: MARGINALLY SUITABLE LONG-TERM
```

```

if ~isempty(coreComments)
    fprintf('- Notes:\n');
    for i = 1:numel(coreComments)
        fprintf('  * %s\n', coreComments(i));
    end
end

fprintf('==================================================\n');
```

```
==============================================
```

```
%% local functions
function x = toNum(s)
    s = strtrim(string(s));

    if s == "NA" || s == ""
        x = NaN;
    else
        x = str2double(s);
    end
end

function out = classifyLight(v, darkTh, brightTh)
    if isnan(v)
        out = "NA";
    elseif v <= darkTh
        out = "DARK";
    elseif v < brightTh
        out = "NORMAL";
    else
        out = "BRIGHT";
    end
end

function [isPass, closeness, centerOffsetPercent] = compareToRange(v, vMin, vMax)
    if isnan(v) || isnan(vMin) || isnan(vMax) || (vMax <= vMin)
        isPass = false;
        closeness = NaN;
        centerOffsetPercent = NaN;
        return
    end

    center = (vMin + vMax) / 2;
    halfRange = (vMax - vMin) / 2;

    centerOffsetPercent = abs(v - center) / halfRange * 100;
    closeness = max(0, 100 - centerOffsetPercent);
    isPass = (v >= vMin) && (v <= vMax);
end

function tr = analyzeLongTermTrend(tHr, y, yMin, yMax, maxAllowedDrift,
minDirectionalFraction, windowFraction)
    good = ~(isnan(tHr) | isnan(y));
    tHr = tHr(good);
    y = y(good);

    tr.startEndDrift = NaN;
    tr.directionalFraction = NaN;
    tr.slopePerHour = NaN;
    tr.recentSlopePerHour = NaN;
    tr.timeToBoundaryHours = Inf;
```

```
    tr.behavior = "INSUFFICIENT_DATA";
    tr.trendingTowardLimit = false;
    tr.predictedEndValue24h = NaN;
    tr.startMean = NaN;
    tr.endMean = NaN;
    tr.settlingLike = false;

    if numel(y) < 5
        return
    end

    n = numel(y);
    w = max(2, round(windowFraction * n));

    yStart = mean(y(1:w), 'omitnan');
    yEnd = mean(y(end-w+1:end), 'omitnan');
    drift = yEnd - yStart;

    tr.startMean = yStart;
    tr.endMean = yEnd;
    tr.startEndDrift = drift;

    dy = diff(y);

    if drift > 0
        dirFrac = mean(dy > 0, 'omitnan');
    elseif drift < 0
        dirFrac = mean(dy < 0, 'omitnan');
    else
        dirFrac = 0.5;
    end

    tr.directionalFraction = dirFrac;

    pAll = polyfit(tHr, y, 1);
    tr.slopePerHour = pAll(1);

    recentStartIdx = max(1, n - max(3, round(0.30 * n)) + 1);
    tRecent = tHr(recentStartIdx:end);
    yRecent = y(recentStartIdx:end);

    if numel(yRecent) >= 3
        pRecent = polyfit(tRecent, yRecent, 1);
        tr.recentSlopePerHour = pRecent(1);
    end

    yNow = y(end);

    if ~isnan(tr.recentSlopePerHour)
        tr.predictedEndValue24h = yNow + tr.recentSlopePerHour * 24;
```

```
    end

    slopeSettling = false;
    if ~isnan(tr.slopePerHour) && ~isnan(tr.recentSlopePerHour) &&
abs(tr.slopePerHour) > eps
        slopeSettling = abs(tr.recentSlopePerHour) <= 0.35 * abs(tr.slopePerHour);
    end

    returningTowardStart = false;
    if ~isnan(yStart)
        returningTowardStart = abs(yNow - yStart) < abs(yEnd - yStart) + 1e-9;
    end

    flattening = false;
    recentHalf = floor(numel(yRecent)/2);
    if recentHalf >= 2
        p1 = polyfit(tRecent(1:recentHalf), yRecent(1:recentHalf), 1);
        p2 = polyfit(tRecent(recentHalf:end), yRecent(recentHalf:end), 1);
        flattening = abs(p2(1)) < abs(p1(1));
    end

    tr.settlingLike = slopeSettling || returningTowardStart || flattening;

    if abs(drift) <= maxAllowedDrift
        tr.behavior = "STABLE";

    elseif dirFrac < minDirectionalFraction
        tr.behavior = "TEMPORARY_BUMP";

    elseif tr.settlingLike
        tr.behavior = "SETTLING";

    else
        slopeToUse = tr.recentSlopePerHour;
        if isnan(slopeToUse)
            slopeToUse = tr.slopePerHour;
        end

        if slopeToUse > 0
            tr.trendingTowardLimit = yNow < yMax;
            if tr.trendingTowardLimit
                dt = (yMax - yNow) / slopeToUse;
                if dt > 0
                    tr.timeToBoundaryHours = dt;
                end
            end
        elseif slopeToUse < 0
            tr.trendingTowardLimit = yNow > yMin;
            if tr.trendingTowardLimit
                dt = (yMin - yNow) / slopeToUse;
```

```
                if dt > 0
                    tr.timeToBoundaryHours = dt;
                end
            end
        end

        if isfinite(tr.timeToBoundaryHours) && tr.timeToBoundaryHours <= 24
            tr.behavior = "UNSAFE_RISING";
        elseif isfinite(tr.timeToBoundaryHours) && tr.timeToBoundaryHours <= 168
            tr.behavior = "WATCH";
        else
            tr.behavior = "SLOW_DRIFT";
        end
    end

    if any(strcmp(string(tr.behavior), ["STABLE","TEMPORARY_BUMP","SETTLING"]))
        tr.timeToBoundaryHours = Inf;
        tr.trendingTowardLimit = false;
    end
end

function [penalty, comments] = applyRealDriftPenalty(penalty, comments, name, tr,
~, watchHours)
    switch string(tr.behavior)
        case "STABLE"

        case "TEMPORARY_BUMP"
            comments(end+1,1) = name + " changed briefly, but not enough to count
as real drift.";

        case "SETTLING"
            comments(end+1,1) = name + " looks like it is settling, so no
continuing boundary estimate is needed.";

        case "SLOW_DRIFT"
            if isfinite(tr.timeToBoundaryHours) && tr.timeToBoundaryHours >
watchHours
                comments(end+1,1) = name + " is drifting, but the boundary looks
far away, so near-future risk is low.";
            else
                penalty = penalty + 8;
                comments(end+1,1) = name + " shows drift and should be watched.";
            end

        case "WATCH"
            penalty = penalty + 15;
            comments(end+1,1) = name + " is trending enough to affect the near
future.";

        case "UNSAFE_RISING"
```

```
            penalty = penalty + 30;
            comments(end+1,1) = name + " is changing consistently enough to become
unsafe soon.";

        otherwise
            penalty = penalty + 5;
            comments(end+1,1) = name + " did not have enough clear data for a
strong prediction.";
    end
end

function txt = passText(tf)
    if tf
        txt = "PASS";
    else
        txt = "FAIL";
    end
end

function printTrend(name, tr)
    fprintf('- %s: ', name);

    if isnan(tr.startEndDrift)
        fprintf('insufficient data\n');
        return
    end

    fprintf('drift = %.3f, directional consistency = %.2f, behavior = %s', ...
        tr.startEndDrift, tr.directionalFraction, tr.behavior);

    if isfinite(tr.timeToBoundaryHours)
        fprintf(' | est. time to boundary = %.2f hr\n', tr.timeToBoundaryHours);
    else
        fprintf(' | no boundary estimate needed\n');
    end
end

function v = displayBoundaryTime(tr)
    if isfinite(tr.timeToBoundaryHours)
        v = tr.timeToBoundaryHours;
    else
        v = NaN;
    end
end

function energyWh = integratePowerWh(tSec, pW)
    good = ~(isnan(tSec) | isnan(pW));
    tSec = tSec(good);
    pW = pW(good);
```

```
    if numel(tSec) < 2
        energyWh = NaN;
        return
    end

    energyWh = trapz(tSec / 3600, pW);
end

function yPred = predictValueAtHours(tSec, y, expectedMin, expectedMax, tr, cfg,
isEnergyVariable, hoursAhead)
    pred = buildPredictionPath(tSec, y, expectedMin, expectedMax, tr, cfg,
isEnergyVariable);

    if isempty(pred.t)
        yPred = NaN;
        return
    end

    tTarget = max(tSec) + hoursAhead * 3600;

    if tTarget < pred.t(1) || tTarget > pred.t(end)
        yPred = NaN;
        return
    end

    yPred = interp1(pred.t, pred.y, tTarget, 'linear', NaN);
end

function pred = buildPredictionPath(tSec, y, expectedMin, expectedMax, tr, cfg,
isEnergyVariable)
    pred.t = [];
    pred.y = [];
    pred.mode = "none";

    good = ~(isnan(tSec) | isnan(y));
    tSec = tSec(good);
    y = y(good);

    if numel(y) < max(6, cfg.minPointsRecent)
        return
    end

    n = numel(y);
    nRecent = max(cfg.minPointsRecent, round(cfg.recentFrac * n));
    nStart = max(3, round(cfg.startFrac * n));

    iRecent = max(1, n - nRecent + 1);

    tRecent = tSec(iRecent:end);
    yRecent = y(iRecent:end);
```

```
    startMean = mean(y(1:nStart), 'omitnan');
    recentMean = mean(yRecent, 'omitnan');
    lastVal = y(end);

    tRecentHr = (tRecent - tRecent(1)) / 3600;
    tLast = tSec(end);

    pRecent = polyfit(tRecent/3600, yRecent, 1);
    recentSlope = pRecent(1);

    pAll = polyfit(tSec/3600, y, 1);
    fullSlope = pAll(1);

    rangeWidth = expectedMax - expectedMin;
    if rangeWidth <= 0
        rangeWidth = max(abs(lastVal), 1);
    end

    if isEnergyVariable
        settleSlopeFracToUse = cfg.energySettleSlopeFrac;
        minSlopeForLinearToUse = cfg.energyMinSlopeForLinear;
    else
        settleSlopeFracToUse = cfg.settleSlopeFrac;
        minSlopeForLinearToUse = max(0.004, cfg.minSlopeForLinear * 0.25);
    end

    slopeSettling = false;
    if ~isnan(fullSlope) && ~isnan(recentSlope) && abs(fullSlope) > eps
        slopeSettling = abs(recentSlope) <= settleSlopeFracToUse * abs(fullSlope);
    end

    returningTowardStart = false;
    if ~isnan(startMean)
        distNow = abs(lastVal - startMean);
        distRecentMean = abs(recentMean - startMean);
        returningTowardStart = distNow <= distRecentMean + cfg.returnFrac *
rangeWidth;
    end

    recentHalf = floor(numel(yRecent)/2);
    flattening = false;
    if recentHalf >= 2
        p1 = polyfit(tRecent(1:recentHalf)/3600, yRecent(1:recentHalf), 1);
        p2 = polyfit(tRecent(recentHalf:end)/3600, yRecent(recentHalf:end), 1);
        flattening = abs(p2(1)) < abs(p1(1));
    end

    strongRecentDrift = abs(recentSlope) >= minSlopeForLinearToUse;
    sameDirection = sign(recentSlope) == sign(fullSlope) || abs(fullSlope) < eps;
```

```
    if isEnergyVariable
        shouldSettle = false;
    else
        shouldSettle = ~strongRecentDrift && ...
            (slopeSettling || returningTowardStart || flattening || ...
             strcmp(string(tr.behavior),"TEMPORARY_BUMP") || ...
             strcmp(string(tr.behavior),"STABLE") || ...
             strcmp(string(tr.behavior),"SETTLING"));
    end

    if strongRecentDrift && sameDirection
        shouldSettle = false;
    end

    if shouldSettle
        target = recentMean;
        if abs(startMean - lastVal) < abs(recentMean - lastVal)
            target = startMean;
        end

        target = min(max(target, expectedMin), expectedMax);

        tFutureEnd = tLast + min(cfg.maxSettleHours, cfg.graphMaxHours) * 3600;
        tFuture = linspace(tLast, tFutureEnd, cfg.curvePoints);

        tau = max((min(cfg.maxSettleHours, cfg.graphMaxHours) * 3600)/4, 60);
        yFuture = target + (lastVal - target) * exp(-(tFuture - tLast)/tau);

        pred.mode = "settling";
        pred.t = tFuture;
        pred.y = yFuture;
        return
    end

    curveUsed = false;
    if cfg.allowCurved && numel(yRecent) >= 5
        pLinRecent = polyfit(tRecentHr, yRecent, 1);
        yLinRecent = polyval(pLinRecent, tRecentHr);

        pQuadRecent = polyfit(tRecentHr, yRecent, 2);
        yQuadRecent = polyval(pQuadRecent, tRecentHr);

        rmseLin = sqrt(mean((yRecent - yLinRecent).^2, 'omitnan'));
        rmseQuad = sqrt(mean((yRecent - yQuadRecent).^2, 'omitnan'));

        curveStrength = abs(2 * pQuadRecent(1));
        slopeAtEnd = 2 * pQuadRecent(1) * tRecentHr(end) + pQuadRecent(2);

        dataSpanHr = max(tRecentHr) - min(tRecentHr);
```

```
        if dataSpanHr <= 0
            dataSpanHr = 1;
        end

        maxAllowedAccel = cfg.maxCurveAccelFrac * max(abs(recentSlope),
minSlopeForLinearToUse) / dataSpanHr;
        curveAccel = min(max(2 * pQuadRecent(1), -maxAllowedAccel),
maxAllowedAccel);

        sameDirectionAsRecent = true;
        if abs(recentSlope) > eps && abs(slopeAtEnd) > eps
            sameDirectionAsRecent = sign(slopeAtEnd) == sign(recentSlope);
        end

        curveOkay = (rmseQuad <= cfg.curveImproveFrac * rmseLin || rmseLin < eps)
&& ...
                    (abs(curveStrength) >= cfg.minCurveStrength || abs(recentSlope)
>= minSlopeForLinearToUse) && ...
                    sameDirectionAsRecent;

        if curveOkay
            maxFutureSec = min(cfg.maxLinearHours, cfg.graphMaxHours) * 3600;
            tFuture = linspace(tLast, tLast + maxFutureSec, cfg.curvePoints);
            dtHr = (tFuture - tLast) / 3600;
            yFuture = lastVal + slopeAtEnd * dtHr + 0.5 * curveAccel * dtHr.^2;

            idxStop = firstBoundaryHit(yFuture, expectedMin, expectedMax);
            if ~isempty(idxStop)
                tFuture = tFuture(1:idxStop);
                yFuture = yFuture(1:idxStop);
                yFuture(end) = min(max(yFuture(end), expectedMin), expectedMax);
            end

            if numel(yFuture) >= 2
                pred.mode = "curved";
                pred.t = tFuture;
                pred.y = yFuture;
                curveUsed = true;
            end
        end
    end

    if curveUsed
        return
    end

    maxFutureSec = min(cfg.maxLinearHours, cfg.graphMaxHours) * 3600;
    tBoundary = NaN;

    if recentSlope > 0 && lastVal < expectedMax
```

```
        dtHr = (expectedMax - lastVal) / recentSlope;
        if dtHr > 0
            tBoundary = tLast + dtHr * 3600;
        end
    elseif recentSlope < 0 && lastVal > expectedMin
        dtHr = (expectedMin - lastVal) / recentSlope;
        if dtHr > 0
            tBoundary = tLast + dtHr * 3600;
        end
    end

    if isnan(tBoundary)
        tEnd = tLast + maxFutureSec;
    else
        tEnd = min(tBoundary, tLast + maxFutureSec);
    end

    if tEnd <= tLast
        return
    end

    tFuture = linspace(tLast, tEnd, cfg.curvePoints);
    yFuture = lastVal + recentSlope * ((tFuture - tLast)/3600);

    if ~isnan(tBoundary) && abs(tFuture(end) - tBoundary) < 1
        if recentSlope > 0
            yFuture(end) = expectedMax;
        elseif recentSlope < 0
            yFuture(end) = expectedMin;
        end
    end

    pred.mode = "linear";
    pred.t = tFuture;
    pred.y = yFuture;
end

function idx = firstBoundaryHit(y, yMin, yMax)
    idx = find(y <= yMin | y >= yMax, 1, 'first');
end

function plotVariableWithBand(t, yMin, yMax, yMid, expectedMin, expectedMax,
plotTitleText, yLabelText, runNum, showBands, showTrend, tr, predictionCfg,
predictionMinExtraFrac, predictionMaxExtraFrac, isEnergyVariable)
    figure
    hold on

    good = ~(isnan(t) | isnan(yMid));

    if nnz(good) < 2
```

```
        if showBands
            plot(t, yMin, '--', 'LineWidth', 1)
            plot(t, yMax, '--', 'LineWidth', 1)
        end

        plot(t, yMid, 'LineWidth', 2)
        yline(expectedMin, '--', 'Expected Min')
        yline(expectedMax, '--', 'Expected Max')
        grid on
        xlabel('Time (s)')
        ylabel(yLabelText)
        title("RUN" + runNum + " - " + plotTitleText)
        legend('Location','best')
        return
    end

    tData = t(good);
    yData = yMid(good);

    if showBands
        plot(t, yMin, '--', 'LineWidth', 1)
        plot(t, yMax, '--', 'LineWidth', 1)
    end

    plot(t, yMid, 'LineWidth', 2)
    yline(expectedMin, '--', 'Expected Min')
    yline(expectedMax, '--', 'Expected Max')

    if showTrend
        p = polyfit(tData, yData, 1);
        yFit = polyval(p, tData);
        plot(tData, yFit, ':', 'LineWidth', 2)
    end

    pred = buildPredictionPath(tData, yData, expectedMin, expectedMax, tr,
predictionCfg, isEnergyVariable);

    predictionPlotted = false;
    xEndPlot = max(tData);

    if ~isempty(pred.t)
        plot(pred.t, pred.y, '-.', 'LineWidth', 1.8)
        predictionPlotted = true;
        xEndPlot = min(max(pred.t), max(tData) + predictionCfg.graphMaxHours *
3600);
    end

    runDuration = max(tData) - min(tData);
    minExtra = predictionMinExtraFrac * max(runDuration, 1);
```

```
    maxExtra = min(predictionMaxExtraFrac * max(runDuration, 1),
predictionCfg.graphMaxHours * 3600);

    desiredExtra = xEndPlot - max(tData);
    desiredExtra = max(desiredExtra, minExtra);
    desiredExtra = min(desiredExtra, maxExtra);

    xlim([min(tData), max(tData) + desiredExtra])

    grid on
    xlabel('Time (s)')
    ylabel(yLabelText)
    title("RUN" + runNum + " - " + plotTitleText)

    if predictionPlotted && showBands && showTrend
        legend('Min','Max','Mid','Expected Min','Expected
Max','Trend','Prediction','Location','best')
    elseif predictionPlotted && showBands
        legend('Min','Max','Mid','Expected Min','Expected
Max','Prediction','Location','best')
    elseif predictionPlotted && showTrend
        legend('Mid','Expected Min','Expected
Max','Trend','Prediction','Location','best')
    elseif predictionPlotted
        legend('Mid','Expected Min','Expected Max','Prediction','Location','best')
    elseif showBands && showTrend
        legend('Min','Max','Mid','Expected Min','Expected
Max','Trend','Location','best')
    elseif showBands
        legend('Min','Max','Mid','Expected Min','Expected Max','Location','best')
    elseif showTrend
        legend('Mid','Expected Min','Expected Max','Trend','Location','best')
    else
        legend('Mid','Expected Min','Expected Max','Location','best')
    end
end
```